REVA UNIVERSITY • SCHOOL OF COMPUTING & INFORMATION TECHNOLOGY

AIML-401 • OFFICIAL FACULTY LECTURE MATERIAL

Course Material: Generative Adversarial Networks (GANs) & Minimax Game

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1. Primary Teaching Points & Concept Excerpt

“Generative Adversarial Networks (GANs) formulate generative modeling as a minimax two-player game between a Generator network G(z) capturing the empirical data distribution and a Discriminator network D(x) estimating probability that a sample originated from the training dataset rather than G.”

2. Mathematical Formulation & Core Equations

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1. Minimax Value Function Formulation:
    min_G max_D V(D, G) = E_{x~p_data}[log D(x)] + E_{z~p_z}[log(1 - D(G(z)))]

2. Optimal Discriminator Closed-Form Solution:
    D*(x) = p_data(x) / (p_data(x) + p_g(x))

3. Jensen-Shannon Divergence Connection:
    V(D*, G) = -log(4) + 2 • JSD(p_data || p_g)

4. Wasserstein GAN Earth-Mover Distance:
    W(p_data, p_g) = sup_{||f||_L ≤ 1} E_{x~p_data}[f(x)] - E_{x~p_g}[f(x)]
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3. 4-Stage Algorithmic Architecture Pipeline

Stage	Pipeline Component	Mathematical / Architectural Operation	Key Objective
1	Latent Sampling	Draw latent noise vectors z ~ N(0, I) from Gaussian prior	Stochastic input seed
2	Generator Synthesis	Pass z through transpose convolutions to synthesize G(z)	Synthetic sample generation
3	Discriminator Scoring	Evaluate D(x_real) and D(G(z)), compute binary cross-entropy	Adversarial discrimination
4	Alternating Backprop	Update Discriminator weights, then update Generator via max log D(G(z))	Nash equilibrium pursuit

4. High-Yield Question & Marking Rubric for Midterm IA-2 & SEE Exams

Expected University Exam Question (8 Marks): *'State the minimax objective function of GANs. Prove that the theoretical global optimum is achieved if and only if p_g = p_data, and calculate the value of V(D*, G) at this optimum point.'*

Official Faculty Marking Rubric:

• 2 Marks: Formulating the full expectation objective function V(D, G).

• 3 Marks: Calculus derivation of the optimal discriminator D*(x).

• 3 Marks: Substituting D*(x) back into V to prove connection to Jensen-Shannon Divergence.